

Linux runs on a range of hardware, right from supercomputers to watches. You can give new life to your old and slow Windows system by installing a lightweight Linux system, or even run a NAS or media streamer using a particular distribution of Linux. There are also lightweight desktops like Xfce and LXDE which can run on lower-end PCs.

5. **Free:** Linux is completely free and users do not need to pay for anything. All the basic software required by a typical user and even an advanced user are available. Dozens of educational software are available under Linux. Even the equivalent of professional software for desktop publishing, photo editing, audio editing and video editing are available. Businesses can use the software free of cost and reduce their IT budgets substantially.
6. **Open source:** The most important aspect of Linux is that its source code is available as it falls under the FOSS category (Free and Open Source Software). The developer community benefits from this as its members have the freedom to view and modify the source code, which is not the case with proprietary software, which they can only use. Several countries are developing their own versions of Linux. This will ultimately help these countries in developing their own OSs for specialised or strategic areas like defence, communications, etc., to begin with. Cyber warfare will play an important role in any war in the future. Usage of foreign and compromised computer operating systems and communication software can bring the country to its knees within a few days, both militarily and economically. Hence, using FOSS tools is critical for the security of any country. Several countries including China, Russia and Cuba are developing their own OSs based on Linux. C-DAC has developed BOSS (Bharat Operating System Solutions), India's own PC operating system.
7. **Ease of use:** Contrary to the general belief that Linux is only for geeks, it has now become user-friendly and has a good graphical user interface (GUI). It has almost all the functionality that Windows has. The GUI has developed to the extent that most of what typical users want can be done on Linux, as easily as it is done in Windows, without knowing any commands.

In case you are using some applications which run only on Windows, you can install Wine (Windows Emulator), using which you can run those applications on a Linux system.

Though the perception is that Linux is not gamer friendly, several games are now available on Linux. Even if a particular game is not available on Linux, you can install *PlayOnLinux* to run Windows games.

8. **Customisation:** Users have tremendous flexibility in customising the system as per their requirements. There are numerous choices for wallpapers, desktop icons and panels. There are more than half-a-dozen desktop environments to choose from, like GNOME, KDE, etc. For any task, right from the GUI interface and file managers, to DVD burners and browsers, around four

to six options are available for any particular software. The Linux versions of most popular browsers are available. The Linux philosophy is based on using several small programs, each of which does one task very well. But these programs can be combined to write really powerful programs and utilities. The Linux OS offers a command line interface with several shells to choose from. Systems administrators can enjoy the powerful command line interface and write shell scripts to automate routine maintenance and various other tasks.

9. **Education:** This is the most useful aspect for students, as they can use the software to study how it works, before modifying and extending the code to suit their needs. This will also help them to learn the internals of an OS and the software. This process will help in the development of new software and aid innovation based on local needs. Even if users are not programmers, they can contribute to Linux by helping in documentation, translation and testing.

Linux can be a fantastic educational tool for schools and colleges as free software is available to aid in teaching. Proprietary software for computation, like MATLAB, is very expensive. There are alternatives available to it like Scilab and GNU Octave. Linux software is available in many areas—Celestia and Stellarium for astronomy, Avogadro and Gabsedit for chemistry, EMOSS and TreeView X for biology; and ROOT, Octopus and Step for physics. You can access the link <https://linuxmint.tumblr.com/post/27284846595/92-of-the-best-free-scientific-software-for-linux> to get a comprehensive list of scientific software for Linux.

10. **Support:** There is strong community support for Linux over the Internet through various forums. Any question posed in forums will usually get a quick response as a lot of volunteers are online and working to solve the problems due to their passion for Linux. The paid support option is also available for commercial enterprises, with companies like Red Hat and Novell offering 24x7 support for critical applications and services. 

References

- [1] Can the Internet exist without Linux? (<http://www.zdnet.com/article/can-the-internet-exist-without-linux/>)
- [2] <https://linuxmint.tumblr.com/post/27284846595/92-of-the-best-free-scientific-software-for-linux>

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